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My Learning Experience

Before this lab, I knew one mathematical difference between median and mean. Now I understand how to use this math for data preparation. I see that one mean changes too much if data has very old adults or small babies. One median is more safe to clean data.

Also, I analyzed my old work. In one past lab, I used numbers to replace text categories. I created 'Sex_Code' with numbers 0 and 1. I created 'Embarked_Code' with numbers 1, 2, and 3. This method has one big problem. One machine learning model can read these numbers like one mathematical scale. That model can think number 3 is mathematically bigger than number 1. In this lab, I use One-Hot Encoding. This method makes one new column for every category. It uses True and False. This logic stops that math problem. True and False make all categories equal.

I did not delete that 'Cabin' column. This action was not in my lab instruction. Also, in my past work, I extracted deck data from this 'Cabin' text. I only want to note one problem. It is not clear how to fix these missing values or if I need to delete this column later.

Now, I want to run this new clean dataset again in one machine learning model. I want to check if this data preparation makes survival predictions better. This shows that data cleaning is one very big part of real data science work.